

Gerasimos Gerogiannis

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RESEARCH INTERESTS

- Computer architecture and high-performance computing.
- Accelerator-centric heterogeneous architectures for machine learning, graph analytics, and scientific workloads.
- Machine learning and its application to systems and computer architecture.

EDUCATION

University of Illinois at Urbana-Champaign (UIUC)

Urbana, IL, USA

PhD - Electrical and Computer Engineering

Aug. 2021 - present

- Thesis: 'Heterogeneity without the Headache: Architecting Accelerator-Centric Computing Systems'
- Advisor: Prof. Josep Torrellas
- GPA: 4.0/4.0

University of Patras

Patras, Greece

Diploma in Electrical and Computer Engineering (BSc + MSc)

Sep. 2016 - June 2021

- Thesis: 'Reinforcement Learning for Task Offloading in Next Generation Networks: Algorithms and Hardware Acceleration'
- Thesis Supervisor: Prof. Alexios Birbas
- GPA: 9.83/10.00 (first of class)

WORK EXPERIENCE

Intel Corporation

January 2024 - December 2025

CPU Architecture & Performance Intern

(remote) Champaign, IL, USA

- CPU architecture redesign for emerging machine learning applications such as large language models.
- Research conducted in internship led to publications in MICRO'25 and ISCA'26 and 4 filed US patents.

Intel Corporation

May 2023 - August 2023

Accelerator Architecture & Performance Intern

(remote) Champaign, IL, USA

- Heterogeneous accelerator architectures to accelerate sparse matrix - dense matrix multiplication.
- Research conducted in internship led to publication in HPCA'24.

Intel Corporation

May 2022 - August 2022

Accelerator Architecture & Performance Intern

(remote) Champaign, IL, USA

- Synergies between hardware and software to maximize performance on the Intel PIUMA accelerator.
- Research conducted in internship led to publication in ISPASS'23.

i-acoma group, University of Illinois at Urbana-Champaign

August 2021 - present

Research Assistant

Champaign, IL, USA

AWARDS AND HONORS

MICRO PhD Forum

IEEE TCuARCH, 2025

- Selected to present my PhD research at the PhD Forum of the International Symposium on Microarchitecture.

Wen-mei W. Hwu Award

UIUC, ECE Department, 2025

- For demonstrating research expertise in the area of Computer Engineering.

Mavis Future Faculty Fellowship

UIUC, Grainger College of Engineering, 2025

- Awarded to facilitate the training of the next generation of engineering faculty.

Rambus Computer Engineering Fellowship

UIUC, ECE Department, 2024

- For demonstrating outstanding research performance in the area of Computer Engineering.

- IEEE Micro Top Picks from Computer Architecture Conferences Top Pick** IEEE, 2024
- Awarded to the 12 most significant papers in computer architecture published in the previous year based on novelty and potential for long-term impact. Top Pick awarded for paper: 'Micro-Armed Bandit: Lightweight & Reusable Reinforcement Learning for Microarchitecture Decision-Making'.
- IEEE Micro Top Picks from Computer Architecture Conferences Honorable Mention** IEEE, 2024
- Awarded to the 12 most significant papers in computer architecture published in the previous year based on novelty and potential for long-term impact. Honorable Mention awarded for paper: 'SPADE: A Flexible and Scalable Accelerator for SpMM and SDDMM'.
- Conference Travel Grants for ISCA/MICRO/HPCA/ASPLOS** IEEE/ACM, 2023-2025
- Student Excellence Award** University of Patras, Greece, 2021
- For the highest GPA among all 2020-2021 graduates of the ECE Department of the University of Patras.
- Excellence Award** State Scholarships Foundation (IKY), Greece, 2021
- For the highest GPA among all 2020-2021 graduates of the ECE Department of the University of Patras.
- Skouras Foundation Award** Skouras Foundation, Greece, 2020
- For the highest GPA among all undergraduate students in the ECE Department of the University of Patras.
- Freshman Excellence Award** University of Patras, Greece, 2016
- For the highest grade in the National University Entrance Exams (Panhellenic Exams) among the freshmen students in the ECE Department of the University of Patras.
- Distinctions in National High-school Student Competitions** 2013-2016
- Distinctions in National Student Physics, Chemistry, and Math Competitions. Auxiliary member of the national team for the International Physics Olympiad.

CONFERENCE PUBLICATIONS

- [C13] '*ATX: Accelerator Task Extensions*'
Gerasimos Gerogiannis, Stijn Eyerman, Josep Torrellas, and Wim Heirman
 To appear in Proceedings of the International Symposium on Computer Architecture (ISCA), June 2026.
- [C12] '*GRANII: Selection and Ordering of Primitives in GRaph Neural Networks using Input Inspection*'
 Damitha Lenadora, Vimarsh Sathia, **Gerasimos Gerogiannis**, Serif Yesil, Josep Torrellas, and Charith Mendis
 In Proceedings of the International Symposium on Code Generation and Optimization (CGO), February 2026.
- [C11] '*NetSparse: In-Network Acceleration of Distributed Sparse Kernels*'
Gerasimos Gerogiannis, Dimitrios Merkouriadis, Charles Block, Annus Zulfiqar, Filippos Tofalos, Muhammad Shahbaz, and Josep Torrellas
 In Proceedings of the International Symposium on Microarchitecture (MICRO), October 2025.
- [C10] '*DECA: A Near-Core LLM Decompression Accelerator Grounded on a 3D Roofline Model*'
Gerasimos Gerogiannis, Stijn Eyerman, Evangelos Georganas, Wim Heirman, and Josep Torrellas
 In Proceedings of the International Symposium on Microarchitecture (MICRO), October 2025.
- [C9] '*Micro-MAMA: Multi-Agent Reinforcement Learning for Multicore Prefetching*'
 Charles Block, **Gerasimos Gerogiannis** and Josep Torrellas
 In Proceedings of the International Symposium on Microarchitecture (MICRO), October 2025.
- [C8] '*COGNATE: Learning-Based Acceleration of Sparse Tensor Programs on Emerging Hardware*'
 Chamika Sudusinghe, **Gerasimos Gerogiannis**, Damitha Lenadora, Charles Block, Josep Torrellas, and Charith Mendis
 In Proceedings of the International Conference on Machine Learning (ICML), July 2025.
- [C7] '*MeshSlice: Efficient 2D Tensor Parallelism for Distributed DNN Training*'
 Hyounghwook Nam, **Gerasimos Gerogiannis**, and Josep Torrellas
 In Proceedings of the International Symposium on Computer Architecture (ISCA), June 2025.
- [C6] '*Distributed-Memory Parallel Algorithms for Sparse Matrix and Sparse Tall-and-Skinny Matrix Multiplication*'
 Isuru Ranawaka, Md Taufique Hussain, Charles Block, **Gerasimos Gerogiannis**, Josep Torrellas, and Ariful Azad
 In Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC), November 2024.

- [C5] *'Two-Face: Combining Collective and One-Sided Communication for Efficient Distributed SpMM'*
Charles Block*, **Gerasimos Gerogiannis***, Charith Mendis, Ariful Azad, and Josep Torrellas
*Co-first authors, order is alphabetical.
In Proceedings of the International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), April 2024.
- [C4] *'HotTiles: Accelerating SpMM with Heterogeneous Accelerator Architectures'*
Gerasimos Gerogiannis, Sriram Aananthakrishnan, Josep Torrellas, and Ibrahim Hur
In Proceedings of the International Symposium on High Performance Computer Architecture (HPCA), March 2024.
- [C3] *'Micro-Armed Bandit: Lightweight & Reusable Reinforcement Learning for Microarchitecture Decision-Making'*
Gerasimos Gerogiannis and Josep Torrellas
In Proceedings of the International Symposium on Microarchitecture (MICRO), October 2023.
Selected as a Top Pick in IEEE Micro Top Picks from Computer Architecture Conferences.
- [C2] *'SPADE: A Flexible and Scalable Accelerator for SpMM and SDDMM'*
Gerasimos Gerogiannis, Serif Yesil, Damitha Lenadora, Dingyuan Cao, Charith Mendis, and Josep Torrellas
In Proceedings of the International Symposium on Computer Architecture (ISCA), June 2023.
Selected as a Honorable Mention in IEEE Micro Top Picks from Computer Architecture Conferences.
- [C1] *'Characterizing the Scalability of Graph Convolutional Networks on Intel® PIUMA'*
Matthew Adiletta, Jesmin Jahan Tithi, Emmanouil-Ioannis Farsarakis, **Gerasimos Gerogiannis**, Robert Adolf, Robert Benke, Sidharth Kashyap, Samuel Hsia, Kartik Lakhotia, Fabrizio Petrini, Gu-Yeon Wei, and David Brooks
In Proceedings of the International Symposium on Performance Analysis of Systems and Software (ISPASS), April 2023.

JOURNAL PUBLICATIONS

- [J2] *'Practical Online Reinforcement Learning for Microprocessors with Micro-Armed Bandit'*
Gerasimos Gerogiannis and Josep Torrellas
IEEE Micro Magazine, Top Picks in Computer Architecture Special Issue, July-August 2024.
- [J1] *'Deep Reinforcement Learning Acceleration for Real-Time Edge Computing Mixed Integer Programming Problems'*
Gerasimos Gerogiannis, Michael Birbas, Aimilios Leftheriotis, Eleftherios Mylonas, Nikolaos Tzanis, and Alexios Birbas
IEEE Access, January 2022.

WORKSHOP PUBLICATIONS (WITH PROCEEDINGS)

- [W2] *'Performance-Driven Composite Prefetching with Bandits'*
Charles Block, Pedro Palacios Almendros, Abraham Farrell, **Gerasimos Gerogiannis**, and Josep Torrellas
In Proceedings of the 4th Data Prefetching Championship (DPC4) at HPCA 2026.
- [W1] *'Automated Data Selection for Efficient Cost Model Training to Optimize Sparse Matrix Kernels on Emerging Hardware Accelerators'*
Chamika Sudusinghe, **Gerasimos Gerogiannis**, Damitha Lenadora, Charles Block, Josep Torrellas, and Charith Mendis
In Proceedings of the Exploration in AI Today Workshop at ICML 2025.

US PATENTS (FILED)

- [P4] *'Unified Transfer Engine for Near-Core Compute Accelerators'*
Gerasimos Gerogiannis, Stijn Eyerman, and Wim Heirman
US Patent App. 19/342,503 January 2026
- [P3] *'Speculative Invocation of Accelerators in Out-of-Order Core Pipelines'*
Gerasimos Gerogiannis, Stijn Eyerman, and Wim Heirman
US Patent App. 19/342,390 January 2026
- [P2] *'Efficient On-chip Memory Bandwidth Configurations and Distribution for Artificial Intelligence (AI) Workloads'*
Wim Heirman, Stijn Eyerman, and **Gerasimos Gerogiannis**

[P1] *'Methods and Apparatus for a Machine Learning Model Decompression Accelerator'*
Gerasimos Gerogiannis, Stijn Eyerman, Wim Heirman, and Evangelos Georganas
US Patent App. 18/927,638 November 2024.

TALKS

[T6] *'Towards Accelerator-Centric Computing'*
9th Computing Systems Research Day, National Technical University of Athens, January 2026

[T5] *'Accelerator-Centric Computing'*
PhD Forum at the 58th International Symposium on Microarchitecture, October 2025.

[T4] *'Sparsity at Scale: Towards Efficient Distributed Sparse Accelerators'*
SRC TECHCON, September 2024.

[T3] *'Micro-Armed Bandit: Lightweight & Reusable Reinforcement Learning for Microarchitecture Decision-Making'*
Intel Archfest, May 2024.
ACE Center for Evolvable Computing Theme Meeting, December 2023.

[T2] *'Domain-Specific Hardware and Software for Mixed Sparse-Dense Algebra at Scale'*
ACE Center for Evolvable Computing Theme Meeting, August 2023.

[T1] *'Accelerators for Irregular Applications'*
IBM-Illinois Discovery Accelerator Institute Annual Meeting, March 2023.

SELECTED PROJECTS

MiniSPADE	2022-2023
Designed the microarchitecture and taped-out a simplified ASIC prototype of the accelerator described in our ISCA'23 paper using TSMC 65nm technology.	

TEACHING EXPERIENCE

Teaching Assistant	ECE Department, UIUC, Spring 2026
Teaching assistant for ECE 445 Senior Design Project.	
Private Tutor	Patras, Greece, 2016 – 2020
Tutor	Prosimo Education Center, Patras, Greece, 2020 – 2021
Gave lectures and tutored undergraduate students in preparation for their exams in over 15 electrical and computer engineering subjects.	

MENTORING EXPERIENCE

Graduate Student Mentoring	UIUC, 2022 – present
<i>Dimitrios Merkouriadis</i>, PhD student at UIUC: In-Network Acceleration for Sparse Kernels, 2024 - present. <i>Filippos Tofalos</i>, PhD student at UIUC: In-Network Acceleration for Sparse Kernels, 2024 - 2025. <i>Chamika Sudusinghe</i>, PhD student at UIUC: Accelerator Performance Modeling and Tuning, 2023 - present. <i>Charles Block</i>, PhD student at UIUC: Algorithms for Distributed Sparse Kernels, 2022 - 2024.	

GRANT WRITING EXPERIENCE

NSF	2023 – 2028
<i>PPoSS: LARGE: General-Purpose Scalable Technologies for Fundamental Graph Problems</i>	Amount: \$5.000.000
- Lead PI: Josep Torrellas, number of PIs: 10 - Assisted in the preparation and writing of the NSF Grant proposal, focusing on hardware support for scalable graph algorithms and on the hardware-software interaction.	

PROFESSIONAL MEMBERSHIPS

Student Member	May 2023 – present
<i>ACM</i>	

SERVICE

Conference Program Committee Member

MICRO 2026

ACM SoCC 2026

Journal Reviewer

ACM Transactions on Architecture and Code Optimization (TACO)

IEEE Computer Architecture Letters (CAL)

IEEE Transactions on Computers (TC)

TECHNICAL SKILLS

Programming Languages: C/C++, Python, Matlab, Shell/Bash scripting, Perl, VHDL, Verilog.

Parallel Programming: OpenMP, MPI, Posix Threads, CUDA.

Frameworks/Libraries: Vivado, Vivado HLS, Cadence EDA tools, TensorFlow, PyTorch, Deep Graph Library.

Microarchitectural Simulators and Tools: SST, Sniper, gem5, ChampSim, Accel-Sim, DRAMSim, CACTI, McPAT.

Development Tools: git, CMake.